065320

Course Name: Probability and Statistics

Course Outcome

CO1- Know the basics of Statistics and apply these techniques in the core course of their study.
CO2- Understand the fundamental concepts of random variables, and their expectations.
CO3- Understand and apply discrete and continuous probability distributions in real-life situations.
CO4- Apply the Central Limit theorem in real life scenarios.
CO5- Understand and model the relationship between two variables.
CO6- Perform the application of statistical inference in decision-making situations.

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End Term Examination, Odd Semester 2025-26
B.Tech (CS/AIML), II year, III Semester
Subject Code & Name- BMAS 0108 & Probability and Statistics

Time: 3 Hours **Maximum Marks: 45**

Instruction for students:

1. Attempt ALL questions.
2. The terms have their usual meanings.

Section – A

Attempt All Questions. 3 X 6 = 18 Marks

lo.	Detail of Question	Marks	CO	BL	KL
1	A call center receives customer calls randomly throughout the day. The probability that any incoming call is a complaint call is 0.15, independent of previous calls. If X is the random variable which represents the number of calls the agent attends until the first complaint call occurs, then a) Find the probability that the first complaint call occurs on the 5th call. b) What is the probability that the agent attends more than 8 calls before receiving the first complaint call?	3	3	E	C
2	Your basketball team is playing a series of 5 games against your opponent. The winner will be the one who wins more games. If your team is much more skilled and has 75% chances of winning, then what is the probability of your team getting 3 wins?	3	3	E	C

3	In the context of hypothesis testing, explain the concepts of the ***null hypothesis*** and the ***alternative hypothesis***.	3	6	R	U
4	Calculate the Karl Pearson's coefficient between x and y for the following information. $n = 10, \sum x = 71, \sum y = 70, \sum x^2 = 555, \sum y^2 = 526, \sum xy = 527.$	3	5	E	C
5	A manufacturer knows that the condensers he makes contain on an average 1% of defectives. He packs them in boxes of 100. What is the probability that a box picked at random will contain 4 or more faulty condensers? Use Poisson distribution.	3	3	E	C
6	A researcher is testing a new teaching method to determine whether it improves students' average performance compared to the traditional method. The null hypothesis H_0 states that the new method has *no effect* on the average performance, while the alternative hypothesis H_1 states that the new method improves performance. Define **Type I** and **Type II** errors in the context of this hypothesis test.	3	6	R	F

Section – B

*Attempt **All** Questions.* 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
7	A nutritionist claims that a new diet plan increases the average daily protein intake of individuals to at least 60 grams. To verify this claim, she collects a random sample of 15 participants following the diet plan for two weeks. The sample shows a mean protein intake of 56.8 grams with a standard deviation of 7.2 grams. Using a 5% significance level, test whether the average daily protein intake of individuals on this diet plan is significantly less than 60 grams. [Given: The tabulated value of t at 5 % level of significance for 14 degree of freedom = 1.7618]	5	6	An	P

8	An experiment is conducted where 6 fields are planted with corn, three fields of the six having no potassium fertilizer and the other three fields having 80 kgs of potassium fertilizer. The resulting corn yields are shown in the table in bushels per acre: **Field:** 1 | 2 | 3 | 4 | 5 | 6 **Potassium (kg) x:** 0 | 0 | 0 | 80 | 80 | 80 **Corn yield y: (acre)** 120 | 360 | 60 | 1280 | 1120 | 1120 Now, (a) Compute a linear regression equation of y on x. (b) Predict corn yield for a field treated with 60 kgs of fertilizer.	5	5	E	C
9	To test the inoculation against cholera, the following table was obtained: **Attacked** | **Not attacked** **Inoculated** 30 | 160 **Non-inoculated** 140 | 460 Use χ^2-test to defend or refute the statement that the inoculation prevents attack from cholera. (Given: the tabulated value of χ^2 for 1 degree of freedom (d.f.) at 5% level of significance is 3.841.)	5	6	An	P

Section – C

Attempt All Questions. 4 X 3 = 12 Marks

No.	Detail of Question	Marks	CO	BL	KL
10	For a distribution of 250 heights, calculations showed that the mean and standard deviation were 54 inches and 3 inches respectively. It was however, discovered on checking that the two items 64 and 50 in the original data were wrongly written in place of the correct values 62 and 52 inches respectively. Calculate correct mean and standard deviation. **OR,** Below the list of variables (and their possible values). Which of these variables are quantitative and which are qualitative? Out of the four possible scale of data (Nominal, ordinal, interval, ratio), which	4	1	An, R	C, F

	scale does each of them belongs to? Explain your answer. (a) The rating of a movie (b) Birthday (c) Weight (d) Height (e) The total debt of the Italian Government in the year 2010 (f) The Stock Exchange on which a company stocks are traded. (g) Your birth place (h) The letter grades in your mark sheet.				
11	A random variable X has the following probability function: **X = x** | −2 | −1 | 0 | 1 | 2 | 3 **P(x)** | 0.1 | k | 0.2 | 2k | 0.3 | k a) Find the value of **k**. b) Determine $E(X)$, $Variance\ (X)$ of the probability function. c) Also Calculate moment generating function.	4	2	E	C
12	The mean yield for one-acre plot is 662 kilos with a standard deviation 32 kilo. Assuming normal distribution, how many one-acre plots in a batch of 1000 plots would you expect to have yield a) Over 700 kilos? b) Below 650 kilos? Given: $P(0 \leq z \leq 1.19) = 0.3830$ and $P(z \leq -0.38) = 0.35197.$	4	3	E	P